

PERSEC Traffic Accident Risk Prediction Model

Comprehensive Risk Factors for Traffic Accident Prediction

Based on comprehensive research and analysis, the PERSEC application should incorporate the following categories of variables and factors to create an accurate traffic accident risk prediction model:

1. Road and Infrastructure Characteristics

Factor	Description	Data Source	Weight
Road Type	Highway, urban street, crosstown road, etc.	Mapbox + GPS	High
Road Surface Material	Asphalt, concrete, gravel, etc.	Public Works Data	Medium
Road Condition	Potholes, cracks, deterioration levels	Public Works Data + User Reports	High
Road Geometry	Curves, grades, slopes, visibility	Mapbox Elevation Data	High
Lane Configuration	Number of lanes, lane width	Mapbox	Medium
Intersection Type	Signalized, roundabout, stop-controlled	Mapbox + Municipal Data	High
Traffic Control Devices	Signs, signals, markings	Municipal Data	Medium
Speed Limits	Posted speed limits	Mapbox + Municipal Data	High
Horizontal Alignment	Road curvature, banking	Mapbox	Medium
Median Types	Presence and type of medians	Mapbox + Visual Analysis	Medium
Shoulder Width	Width of road shoulders	Mapbox + Visual Analysis	Medium
Roadside Features	Barriers, clear zones, utility poles	Mapbox + Visual Analysis	Medium
Road Network Density	Density of roads in the area	Calculated from Mapbox	Medium
Points of Interest Density	Density of shops, schools, etc.	POI Data API	Medium
Functional Zone Type	Residential, commercial, industrial	Municipal Zoning Data	Medium

2. Environmental and Weather Conditions

Factor	Description	Data Source	Weight
Weather Conditions	Rain, snow, fog, clear	Weather API	High
Road Surface Status	Dry, wet, icy, snowy	Weather API + User Reports	High
Visibility	Clear, limited, poor	Weather API	High
Temperature	Current temperature	Weather API	Medium
Precipitation Intensity	Light, moderate, heavy	Weather API	High
Wind Speed	Current wind speed	Weather API	Medium
Time of Day	Dawn, day, dusk, night	System Clock	High
Lighting Conditions	Daylight, darkness, artificial lighting	Calculated from time + location	High
Sun Position	Glare potential	Calculated from time + location	Medium
Season	Spring, summer, fall, winter	System Calendar	Medium
Air Quality	Smog, haze	Environmental API	Low

3. Traffic Conditions

Factor	Description	Data Source	Weight
Traffic Volume	Vehicles per hour	Traffic API	High
Traffic Density	Vehicles per mile/km	Traffic API	High
Traffic Speed	Average speed of traffic	Traffic API	High
Speed Variance	Variation in vehicle speeds	Calculated from Traffic API	High
Traffic Phase	Free-flowing, congested, etc.	Calculated from Traffic API	High
Historical Accident Frequency	Past accident counts in location	Accident Database	Very High
Traffic Composition	Percentage of trucks, cars, etc.	Traffic API	Medium
Rush Hour Status	Whether current time is rush hour	Calculated from time + location	Medium
Traffic Flow Pattern	Stable, unstable	Calculated from Traffic API	Medium

4. Temporal Patterns

Factor	Description	Data Source	Weight
Time of Day	Hour of day	System Clock	High
Day of Week	Weekday vs weekend	System Calendar	High
Month	Month of year	System Calendar	Medium
Holiday Status	Regular day vs holiday	Holiday Calendar API	Medium
Special Events	Sporting events, concerts, etc.	Events API	Medium
School Session	School in session vs break	School Calendar API	Medium

5. Spatial Context

Factor	Description	Data Source	Weight
Proximity to High-Risk Areas	Distance to known accident hotspots	Calculated from Accident Database	Very High
Urban vs Rural	Area classification	Municipal Data	Medium
Proximity to Schools	Distance to nearest school	POI Data	Medium
Proximity to Entertainment Venues	Distance to bars, clubs, etc.	POI Data	Medium
Proximity to Hospitals	Distance to emergency services	POI Data	Low
Land Use Mix	Diversity of land uses	Municipal Zoning Data	Medium
Population Density	People per square mile/km	Census Data	Medium

6. Visual Semantics

Factor	Description	Data Source	Weight
Visual Complexity	Complex vs simple visual environment	Street View Image Analysis	Medium
Visual Distractions	Billboards, digital signs, etc.	Street View Image Analysis	Medium
Streetscape Elements	Trees, buildings, etc.	Street View Image Analysis	Medium
Road Markings Visibility	Clarity of lane markings	Street View Image Analysis	Medium
Scene Similarity	Similarity to high-risk locations	Calculated from Image Analysis	Medium

7. Human Factors (When Available)

Factor	Description	Data Source	Weight
Driver Age	Age groups of typical drivers	Anonymous Demographics	Medium
Driver Experience	Years of driving experience	Anonymous Demographics	Medium
Population Demographics	Age distribution in area	Census Data	Low
Tourist Area	Tourist vs local population	Tourism Data	Low

Risk Score Calculation Methodology

The PERSEC application will use a hybrid machine learning approach combining multiple models to calculate risk scores:

1. Base Model Architecture

The risk prediction system will use a hybrid model consisting of:

- 1. Random Forest Classifier:** For initial risk categorization
- 2. Gradient Boosting:** For fine-tuning predictions
- 3. Spatial-Temporal Graph Convolutional Network:** For capturing complex spatial and temporal relationships

2. Risk Score Calculation

Each location will receive a normalized risk score from 0-100 based on:

$$\text{Risk Score} = \text{Base Risk Score} \times \text{Temporal Multiplier} \times \text{Weather Multiplier} \times \text{Traffic Multiplier}$$

Where:

- Base Risk Score:** Calculated from static features (road geometry, historical accidents)
- Temporal Multiplier:** Adjusts risk based on time patterns
- Weather Multiplier:** Adjusts risk based on current weather conditions
- Traffic Multiplier:** Adjusts risk based on current traffic conditions

3. Risk Categories

Risk scores will be mapped to five categories:

Risk Score Range	Risk Level	Color Code	User Alert Level
0-20	Minimal Risk	Green	None
21-40	Low Risk	Blue	Informational
41-60	Moderate Risk	Yellow	Awareness
61-80	High Risk	Orange	Caution
81-100	Severe Risk	Red	Warning

Implementation Strategy

1. Data Collection and Preprocessing

1. Establish data pipelines for all required data sources
2. Normalize and standardize data from different sources
3. Implement feature engineering for derived variables
4. Create a unified spatial-temporal framework for all data

2. Model Training

1. Split data into training, validation, and test sets
2. Train base models on historical accident data
3. Implement ensemble methods to combine model outputs
4. Tune hyperparameters using grid search and cross-validation
5. Validate model performance on held-out test data

3. Real-time Prediction System

1. Develop API endpoints for real-time data ingestion
2. Implement streaming data processing for continuous updates
3. Create caching mechanisms for frequently accessed data
4. Establish update frequencies for different data types:
 - Traffic data: 1-5 minute intervals
 - Weather data: 15-30 minute intervals
 - Event data: Daily updates

4. Continuous Improvement

1. Implement A/B testing for model variations
2. Collect user feedback on risk predictions
3. Monitor prediction accuracy against actual accidents
4. Retrain models periodically with new data
5. Perform post-incident analysis to improve predictions

Technical Implementation with Azure and Mapbox

1. Azure Components

- **Azure Cosmos DB:** Store processed risk scores and model outputs

- **Azure SQL Database:** Store historical accident data and static features
- **Azure Functions:** Process real-time data and calculate risk scores
- **Azure Machine Learning:** Train and deploy prediction models
- **Redis Cache:** Cache frequently accessed risk data for quick retrieval

2. Mapbox Integration

- **Mapbox Navigation SDK:** Provide base maps and routing functionality
- **Custom Styling:** Display risk heat maps and route risk visualizations
- **Geocoding API:** Convert locations to/from coordinates
- **Directions API:** Calculate routes with safety prioritization
- **Static Images API:** Retrieve street view images for visual semantic analysis

3. Mobile Implementation

The mobile app will implement:

- Background location tracking with battery optimization
- Local caching of risk data for offline use
- Real-time risk alerts based on location and movement
- Safe route visualization with risk highlighting
- Emergency SOS features tied to risk levels

Model Evaluation Metrics

The prediction model will be evaluated using:

1. **Accuracy:** Overall correctness of risk predictions
2. **Precision:** Proportion of correctly predicted high-risk locations
3. **Recall:** Proportion of actual high-risk locations correctly identified
4. **F1 Score:** Harmonic mean of precision and recall
5. **AUC-ROC:** Area under the receiver operating characteristic curve
6. **Spatial Accuracy:** Correctness of risk predictions within geographic context
7. **Temporal Accuracy:** Correctness of risk predictions over time
8. **False Positive Rate:** Rate of incorrectly identified high-risk locations

Risk Communication to Users

The PERSEC app will communicate risk information to users through:

1. **Map Visualization:** Heat map showing risk levels by color
2. **Route Risk Indicators:** Color-coded segments along navigation routes
3. **Risk Alerts:** Push notifications for entering high-risk areas
4. **Voice Guidance:** Verbal warnings for approaching high-risk areas
5. **Risk Details:** Tap-to-view information about specific risk factors
6. **Safety Recommendations:** Contextual advice based on risk factors

References and Research Basis

This model integrates findings from multiple research studies on traffic accident prediction, including machine learning approaches, spatiotemporal analysis, and risk factor identification. Key influences include Random Forest models for accident severity prediction, SHAP analysis for feature importance interpretation, and hybrid spatiotemporal models that incorporate visual semantics from urban environments.